

LE1_Review_Questions-Periodic_Motion

October 8, 2026

```
[1]: ## Setup!
import math
```

1 Review Questions: Periodic Motion

1. Which statement best defines simple harmonic motion?

“B. The restoring force is directly proportional to the displacement from equilibrium and opposite to it.”. Recall Hooke’s Law, which is the definition of Simple Harmonic Motion (SHM).

$$F = -kx \quad (1)$$

$$a = -\omega^2 x \quad (2)$$

3. An oscillator completes $n = 12$ full cycles in $t = 3.0$ s. What are its frequency f , period t , and angular frequency ω ?

Recall that frequency is the number of cycles per second. So it can be represented by the following formula:

$$f = \frac{n}{t} \quad (3)$$

$$= \frac{12}{3.0 \text{ s}} \quad (4)$$

$$= 4.0 \text{ Hz} \quad (5)$$

```
[2]: n = 12.0 ## full cycles
t = 3.0 ## s
f = n / t ## 1 / s = Hz
print(f"n = {n} full cycles\n"
      f"t = {t:.14g} s\n"
      f"f = {f} Hz")
```

```
n = 12.0 full cycles
t = 3 s
f = 4.0 Hz
```

We can then compute for period T as it is the reciprocal of frequency f .

$$T = \frac{1}{f} \quad (6)$$

$$= \frac{1}{4.0 \text{ Hz}} \quad (7)$$

$$= 0.25 \text{ s} \quad (8)$$

```
[3]: T = 1 / f
      print(f"T = {T} s")
```

T = 0.25 s

Lastly, the angular frequency ω can be derived from another formula for frequency f .

$$(f = \frac{\omega}{2\pi}) \frac{2\pi}{1} \quad (9)$$

$$\omega = 2\pi f \quad (10)$$

$$= 2\pi * 4.0 \text{ Hz} \quad (11)$$

$$= 2\pi * 4.0 \frac{1}{\text{s}} \quad (12)$$

$$= 8\pi \frac{1}{\text{s}} \quad (13)$$

$$= 25.13 \frac{\text{rad}}{\text{s}} \quad (14)$$

```
[4]: ## 2 * math.pi == math.tau
      print(f"math.pi * 2 = {math.pi * 2:.14g}\n"
            f"math.tau      = {math.tau:.14g}")

      omega = math.tau * f
      print(f"omega = {omega} rad / s")
```

math.pi * 2 = 6.2831853071796

math.tau = 6.2831853071796

omega = 25.132741228718345 rad / s

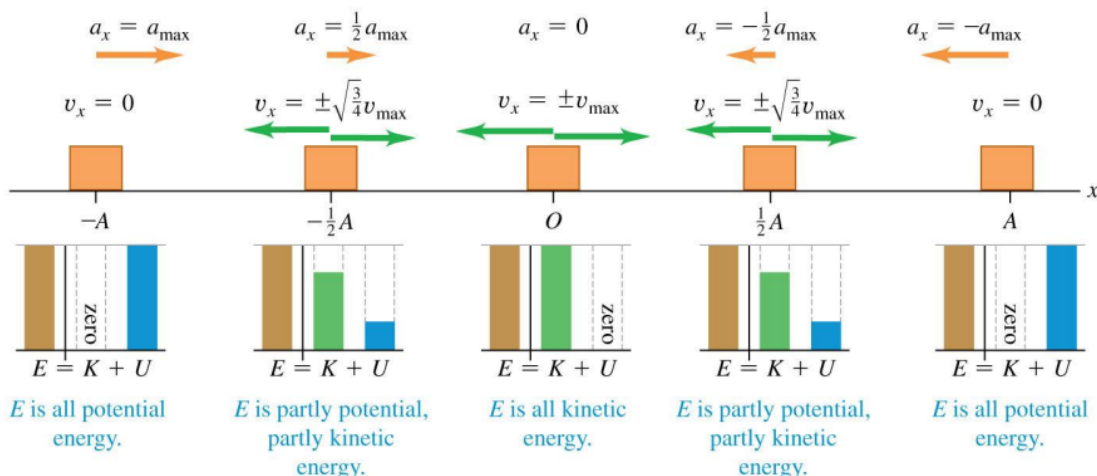
3. The amplitude A of a mass-spring oscillator is doubled (with the same mass m and spring constant k). What happens to the period T , maximum speed v_{max} , and total energy E ?

Recall that in the section *Energy in SHM*, the period T is independent of the amplitude A .

Let $A' = 2A$. Since $v_{max} = \omega A' = \omega 2A$, it doubles. Since $E = \frac{1}{2}k \times A'^2 = \frac{1}{2}k \times 2A'^2$, its exponential nature means a doubling of the amplitude A results in a quadrupling of the total energy E .

- The total mechanical energy $E = K + U$ is conserved in SHM:

$$E = \frac{1}{2} m v_x^2 + \frac{1}{2} k x^2 = \frac{1}{2} k A^2 = \text{constant}$$



Proof that $v_{\max} = \omega A$ when $x = 0$:

$$x(t) = A \cos(\omega t + \phi) \quad (15)$$

$$\frac{0}{A} = \frac{A \cos(\omega t + \phi)}{A} \quad (16)$$

$$0 = \cos(\omega t + \phi) \quad (17)$$

4. A glider on a spring is momentarily at $x = -A$. Which description is correct?

The maximum amplitude A is at either extreme. In this case, the negative extreme (i.e. $-A$) is the leftmost. At this point, the glider has a speed $v_x = 0$ because it is about to switch directions. However, no matter the extreme, the restoring force **always points toward equilibrium**.

5. Several tuning forks have tines of different mass. Which produces the lowest pitch?

Recall the formula for frequency, given some mass m :

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (18)$$

Because the mass m is in the denominator, this means that mass is indirectly proportional to the resulting frequency f . Thus larger masses result in lower frequencies, assuming the same spring constant k .

6. For $x(t) = A \cos(\omega t + \phi)$, which phase angle ϕ describes an object that is at $x = 0$ and $t = 0$ and moving in the $+x$ direction?

Recall the formula for displacement as a function of time for simple harmonic motion, substituting in displacement $x = 0$ m and time $t = 0$ s. We are effectively looking for the phase angle at the **starting point** of the graph.

$$x(t) = A \cos(\omega t + \phi) \quad (19)$$

$$0 = A \cos(\omega \times 0 + \phi) \quad (20)$$

$$0 = A \cos(\phi) \quad (21)$$

$$\text{since:} \quad (22)$$

$$\cos(0) = 1 \quad (23)$$

$$\text{thus:} \quad (24)$$

$$0 = A \cos(0) \quad (25)$$

$$0 = A \times 1 \quad (26)$$

$$0 = A \quad (27)$$

$$\text{thus:} \quad (28)$$

$$\phi = 0 \quad (29)$$

7. A particle's position is $x(t) = (0.30 \text{ m}) \cos(4\pi t + \frac{\pi}{3})$, with t in seconds. What are its period T and maximum speed v_{max} ?

So we are to use the equation for *Position x as a function of time t* , with the following givens:

$$x(t) = (0.30 \text{ m}) \cos(4\pi t + \frac{\pi}{3}) \quad (30)$$

$$A = 0.30 \text{ m} \quad (31)$$

$$\omega = 4\pi \frac{\text{rad}}{\text{s}} \quad (32)$$

$$t = ? \text{ s} \quad (33)$$

$$\phi = \frac{\pi}{3} \text{ rad} \quad (34)$$

```
[5]: A      = 0.30 ## m
     omega = 4 * math.pi ## rad / s
     phi   = math.pi / 3 ## rad

     print(f"A = {A}\nomega = {omega}\nphi = {phi}")
```

A = 0.3

omega = 12.566370614359172

phi = 1.0471975511965976

Since we have ω , we can determine the frequency f :

$$f = \frac{\omega \frac{\text{rad}}{\text{s}}}{2\pi} \quad (35)$$

$$= \frac{4\pi \frac{\text{rad}}{\text{s}}}{2\pi} \quad (36)$$

$$= 2 \frac{1}{\text{s}} \quad (37)$$

$$= 2 \text{ Hz} \quad (38)$$

With the frequency f , we can determine the period T :

$$T = \frac{1}{f} \quad (39)$$

$$= \frac{1}{2 \text{ Hz}} \quad (40)$$

$$= \frac{1}{2 \frac{1}{\text{s}}} \quad (41)$$

$$= \frac{1 \text{ s}}{2} \quad (42)$$

$$= 0.5 \text{ s} \quad (43)$$

To determine the velocity, we need to use the graph of displacement and velocity for simple harmonic motion. The graph states that $v_{\max} = \omega A$. So:

$$v_{\max} = \omega A \quad (44)$$

$$= 4\pi \frac{\text{rad}}{\text{s}} \times 0.30 \text{ m} \quad (45)$$

$$= 3.77 \frac{\text{m}}{\text{s}} \quad (46)$$

```
[6]: v_max = omega * A
      print(f"v_max = {v_max}")
```

```
v_max = 3.7699111843077517
```

8. A block with mass $m = 0.200 \text{ kg}$, on a horizontal spring, with spring constant $k = 80.0 \frac{\text{N}}{\text{m}}$ oscillates with an amplitude $A = 0.100 \text{ m}$. What are its maximum speed and maximum acceleration?

Recall the formulas used for *Energy in Simple Harmonic Motion*. Recall that $v_{\max}$ occurs as displacement $x = 0$. Let's derive the formula for some speed v_x .

$$E = \frac{1}{2}mv_x^2 + \frac{1}{2}kx^2 \quad (47)$$

$$= \frac{1}{2}kA^2 \quad (48)$$

$$\text{thus:} \quad (49)$$

$$(\frac{1}{2}kA^2 = \frac{1}{2}mv_x^2 + \frac{1}{2}kx^2) - (\frac{1}{2}kx^2) \quad (50)$$

$$(\frac{1}{2}kA^2 - \frac{1}{2}kx^2 = \frac{1}{2}mv_x^2) \times \frac{2}{m} \quad (51)$$

$$\frac{2}{m} \times \frac{1}{2}k(A^2 - x^2) = v_x^2 \quad (52)$$

$$v_x^2 = \frac{k}{m}(A^2 - x^2) \quad (53)$$

$$\sqrt{v_x^2} = \sqrt{\frac{k}{m}(A^2 - x^2)} \quad (54)$$

$$v_x = \sqrt{\frac{k}{m}} \times \sqrt{A^2 - x^2} \quad (55)$$

Now let's use the formula for some speed v_x , where $x = 0$ because that is where v_{max} occurs:

$$v_x = \sqrt{\frac{k}{m}} \times \sqrt{A^2 - x^2} \quad (56)$$

$$= \sqrt{\frac{k}{m}} \times \sqrt{A^2 - 0^2} \quad (57)$$

$$= \sqrt{\frac{k}{m}} \times \sqrt{A^2} \quad (58)$$

$$= \sqrt{\frac{k}{m}} \times A \quad (59)$$

$$\text{where:} \quad (60)$$

$$\omega = \sqrt{\frac{k}{m}} \quad (61)$$

$$\text{thus:} \quad (62)$$

$$v_{max} = \omega A \quad (63)$$

So we can solve for the angular frequency ω first and then multiply it to the amplitude A to get the maximum speed v_{max} .

$$\begin{aligned}
 \omega &= \sqrt{\frac{k}{m}} \\
 &= \sqrt{\frac{80.0 \frac{\text{N}}{\text{m}}}{0.200 \text{ kg}}} \\
 &= \sqrt{\frac{80.0 \frac{\text{kg} \cdot \text{m} \cdot \text{s}^{-2}}{\text{m}}}{0.200 \text{ kg}}} \\
 &= \sqrt{\frac{80.0 \text{ s}^{-2}}{0.200}} = \sqrt{\frac{80.0 \frac{1}{\text{s}^2}}{0.200}} \\
 &= \sqrt{400 \frac{1}{\text{s}^2}} \\
 &= 20 \frac{1}{\text{s}} = 20 \frac{\text{rad}}{\text{s}}
 \end{aligned}$$

Then:

$$\begin{aligned}
 v_x &= \omega A \\
 &= 20 \frac{1}{\text{s}} \times 0.100 \text{ m} \\
 &= 2.0 \frac{\text{m}}{\text{s}}
 \end{aligned}$$

So the maximum speed is $v_{max} = 2.0 \frac{\text{m}}{\text{s}}$. To get the maximum acceleration a_{max} , we can use the following formula:

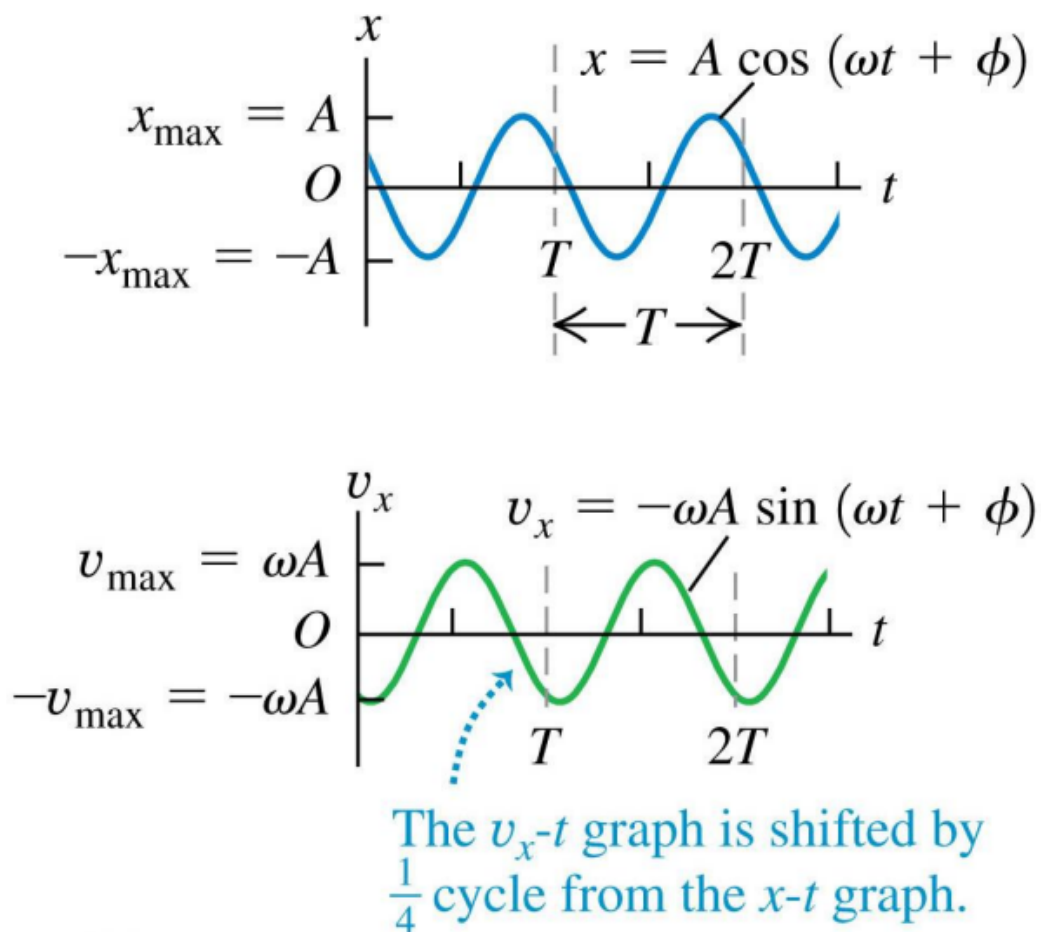
$$\begin{aligned}
 a_{max} &= -\frac{k}{m}(-A) \\
 &= -\frac{80.0 \frac{\text{N}}{\text{m}}}{0.200 \text{ kg}}(-0.100 \text{ m}) \\
 &= -\frac{80.0 \text{ kg} \cdot \text{m} \cdot \text{s}^{-2} \times \frac{1}{\text{m}}}{0.200 \text{ kg}}(-0.100 \text{ m}) \\
 &= -\frac{80.0 \text{ s}^{-2}}{0.200} (-0.100 \text{ m}) \\
 &= 40 \text{ m} \cdot \text{s}^{-2} \\
 &= 40 \frac{\text{m}}{\text{s}^2}
 \end{aligned}$$

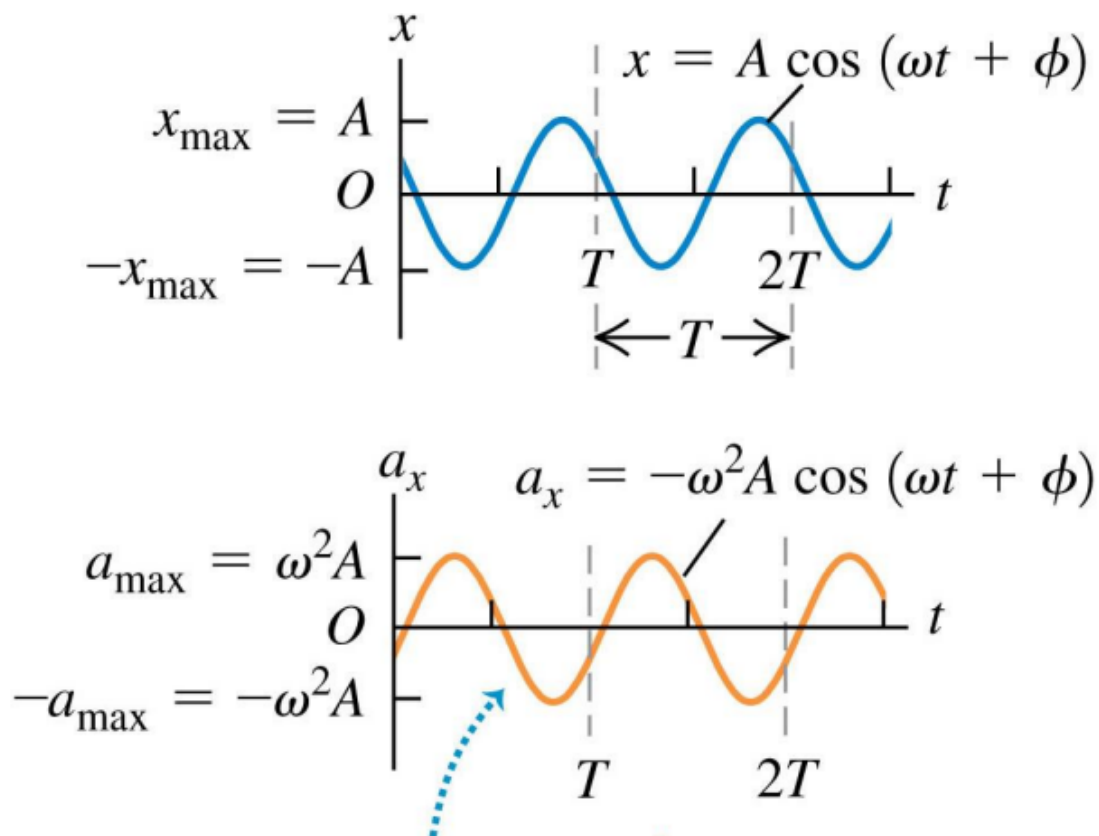
[7]: `-(80.0/.200) * (-.100)`

[7]: 40.0

9. Which statement correctly describes the phase relationship among the displacement as function of time $x(t)$, velocity as a function of time $v(t)$, and amplitude as a function of time $a(t)$ in simple harmonic motion?

Recall the comparisons of the various graphs.





The a_x - t graph is shifted by $\frac{1}{4}$ cycle from the v_x - t graph and by $\frac{1}{2}$ cycle from the x - t graph.

10. A block with mass $m = 0.50$ kg on a spring oscillates at frequency $f = 2.0$ Hz. If it is replaced by another block with new mass $m' = 2.0$ kg on the same spring, what is the new frequency?

First, we need to somehow relate mass and frequency. Recall that for an **angular** frequency ω :

$$\omega = \sqrt{\frac{k}{m}} \quad (64)$$

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \times \sqrt{\frac{k}{m}} \quad (65)$$

$$T = \frac{1}{f} = \frac{2\pi}{\omega} \quad (66)$$

The relation $T = \frac{1}{f}$ is useful because we can directly solve for the time period T and relate it back to get the angular frequency ω .

$$T = \frac{1}{f} \quad (67)$$

$$= \frac{1}{2.0 \frac{1}{\text{s}}} = \frac{1}{2.0} \text{ s} \quad (68)$$

$$= 0.5 \text{ s} \quad (69)$$

$$\text{thus:} \quad (70)$$

$$T = \frac{2\pi}{\omega} \quad (71)$$

$$T \times \frac{\omega}{1} = \frac{2\pi}{\omega} \times \frac{\omega}{1} \quad (72)$$

$$\frac{T\omega}{T} = \frac{2\pi}{T} \quad (73)$$

$$\omega = \frac{2\pi}{T} \quad (74)$$

$$\text{thus:} \quad (75)$$

$$\omega = \frac{2\pi}{0.5 \text{ s}} \quad (76)$$

$$= 12.57 \frac{1}{\text{s}} = 12.57 \frac{\text{rad}}{\text{s}} \quad (77)$$

```
[8]: f = 2.0 ## Hz
     T = 1.0 / f ## s
     omega = math.tau / T ## rad / s

     print(f"f = {f} Hz\nT = {T} s\nomega = {omega} rad / s")
```

```
f = 2.0 Hz
T = 0.5 s
omega = 12.566370614359172 rad / s
```

We can then use the angular frequency f to calculate the spring force constant k . The k for the initial block should be the same as the k for the second one.

$$\omega = \sqrt{\frac{k}{m}} \quad (78)$$

$$\omega = \frac{\sqrt{k}}{\sqrt{m}} \quad (79)$$

$$\omega \times \frac{\sqrt{m}}{1} = \frac{\sqrt{k}}{\sqrt{m}} \times \frac{\sqrt{m}}{1} \quad (80)$$

$$\omega\sqrt{m} = \sqrt{k} \quad (81)$$

$$(\sqrt{k})^2 = (\omega\sqrt{m})^2 \quad (82)$$

$$k = \omega^2 m \quad (83)$$

$$\text{thus:} \quad (84)$$

$$k = (12.57 \frac{\text{rad}}{\text{s}})^2 \times 0.50 \text{ kg} \quad (85)$$

$$k = 157.0049 \frac{1}{\text{s}^2} \times 0.50 \text{ kg} \quad (86)$$

$$k = 79 \text{ kg} \cdot \text{s}^{-2} \quad (87)$$

```
[9]: m = 0.50 ## kg
      k = omega**2 * m ## kg / s^2

      print(f"{omega**2} rad^2 / s^2")
      print(f"m = {m} kg\nk = {k} kg / s^2")
```

157.91367041742973 rad^2 / s^2

m = 0.5 kg

k = 78.95683520871486 kg / s^2

We can now solve for the new frequency f' when a block with some other mass $m' = 2.0 \text{ kg}$.

$$f' = \frac{1}{2\pi} \times \sqrt{\frac{k}{m'}} \quad (88)$$

$$= \frac{1}{2\pi} \times \sqrt{\frac{79 \text{ kg} \cdot \text{m} \cdot \text{s}^{-2}}{2.0 \text{ kg}}} \quad (89)$$

$$= \frac{1}{2\pi} \times \sqrt{39.5 \frac{\text{m}}{\text{s}^2}} \quad (90)$$

$$= 1 \text{ Hz} \quad (91)$$

```
[10]: m = 2.0 ## kg
        print(f"<tmp> = {k / m} m / s^2")

        f = (1 / math.tau) * math.sqrt(k / m)
        print(f"f = {f} Hz")
```

<tmp> = 39.47841760435743 m / s²
 f = 1.0 Hz

11. A mass-spring system has amplitude A . When the mass is at displacement $x = \frac{A}{2}$ what fraction of the total energy is kinetic?

Recall the section *Energy in SHM*.

$$E = K + U \quad (92)$$

$$(-U) + E = K + U + (-U) \quad (93)$$

$$K = E - U \quad (94)$$

$$\text{where:} \quad (95)$$

$$E = \frac{1}{2}kA^2 \quad (96)$$

$$K = \frac{1}{2}mv_x^2 \quad (97)$$

$$U = \frac{1}{2}kx^2 \quad (98)$$

$$\text{thus:} \quad (99)$$

$$\frac{1}{2}kA^2 = \frac{1}{2}mv_x^2 + \frac{1}{2}kx^2 \quad (100)$$

$$\text{where:} \quad (101)$$

$$x = \frac{A}{2} \quad (102)$$

$$\text{thus:} \quad (103)$$

$$K = E - U \quad (104)$$

$$= \frac{1}{2}kA^2 - \frac{1}{2}k\frac{A^2}{4} \quad (105)$$

$$= \frac{1}{2}k\left(A^2 - \frac{A^2}{4}\right) \quad (106)$$

$$= \frac{1}{2}k\left(A^2\frac{4}{4} - \frac{A^2}{4}\right) \quad (107)$$

$$= \frac{1}{2}k\left(\frac{4A^2 - A^2}{4}\right) \quad (108)$$

$$= \frac{1}{2}k\left(\frac{3A^2}{4}\right) \quad (109)$$

$$(110)$$

Thus the factor for kinetic energy K is $\frac{3}{4}$. Note that we don't consider the $\frac{1}{2}$ as part of the factor because that's part of the equation.

12. A spring with mass $m = 0.250$ kg has mass constant $k = 100 \frac{\text{N}}{\text{m}}$ and amplitude $A = 0.20$ m. What is its speed v_x when displacement $x = 0.12$ m?

This is an application of the equation for *Energy*. We have all the parameters sans speed v_x .

$$E = K + U \quad (111)$$

$$K = E - U \quad (112)$$

$$\text{where:} \quad (113)$$

$$E = \frac{1}{2}kA^2 \quad (114)$$

$$K = \frac{1}{2}mv_x^2 \quad (115)$$

$$U = \frac{1}{2}kx^2 \quad (116)$$

$$\text{thus:} \quad (117)$$

$$K = E - U \quad (118)$$

$$\frac{1}{2}mv_x^2 = \frac{1}{2}kA^2 - \frac{1}{2}kx^2 \quad (119)$$

$$\frac{\frac{1}{2}mv_x^2}{\frac{1}{2}m} = \frac{\frac{1}{2}k(A^2 - x^2)}{\frac{1}{2}m} \quad (120)$$

$$v_x^2 = \frac{k(A^2 - x^2)}{m} \quad (121)$$

$$\sqrt{v_x^2} = \sqrt{\frac{k(A^2 - x^2)}{m}} \quad (122)$$

$$v_x = \sqrt{\frac{k(A^2 - x^2)}{m}} \quad (123)$$

$$= \sqrt{\frac{100 \frac{\text{N}}{\text{m}} \times [(0.20 \text{ m})^2 - (0.12 \text{ m})^2]}{0.250 \text{ kg}}} \quad (124)$$

$$= \sqrt{\frac{100 \frac{\text{kg} \cdot \text{m} \cdot \text{s}^{-2}}{\text{m}} \times (0.04 \text{ m}^2 - 0.0144 \text{ m}^2)}{0.250 \text{ kg}}} \quad (125)$$

$$= \sqrt{\frac{100 \text{ s}^{-2} \times 0.0256 \text{ m}^2}{0.250}} \quad (126)$$

$$= \sqrt{\frac{2.56 \text{ m}^2 \cdot \text{s}^{-2}}{0.250}} \quad (127)$$

$$= \sqrt{10.42 \text{ m}^2 \cdot \text{s}^{-2}} \quad (128)$$

$$= 3.2 \text{ m} \cdot \text{s}^{-1} = 3.2 \frac{\text{m}}{\text{s}} \quad (129)$$

```
[11]: # math.sqrt((100 * (.20**2 - .12**2)) / .250)
a, b = .20**2, .12**2
tmp = a - b
print(f"          a^2                = {a:.14g} m^2\n"
```

```

f"          b^2      = {b:.14g} m^2\n"
f"      (a^2 - b^2)  = {tmp:.14g} m^2")

k = 100.0 ## N / m
      ## (kg * m * s^-2) / m
      ## kg / s^-2

tmp = k * tmp ## m / s
print(f"      k(a^2 - b^2)      = {tmp:.14g} m^2 / s^-2")

m   = 0.250 ## kg
tmp = tmp / m
v_x = math.sqrt(tmp)
print(f"      k(a^2 - b^2) / m = {tmp:.14g} m^2 / s^-2\n"
      f"sqrt(k(a^2 - b^2) / m) = {v_x:.14g} m / s")

a^2          = 0.04 m^2
      b^2      = 0.0144 m^2
(a^2 - b^2)   = 0.0256 m^2
k(a^2 - b^2)  = 2.56 m^2 / s^-2
k(a^2 - b^2) / m = 10.24 m^2 / s^-2
sqrt(k(a^2 - b^2) / m) = 3.2 m / s

```

13. A spring has spring force constant $k = 200 \frac{\text{N}}{\text{m}}$ and oscillates with amplitude $A = 0.050 \text{ m}$. What is the kinetic energy at displacement $x = 0.030 \text{ m}$?

The technical energy E is composed of kinetic energy K and potential energy U . They are given by the following formulas:

$$E = K + U \quad (130)$$

$$(-U) + E = K + U + (-U) \quad (131)$$

$$K = E - U \quad (132)$$

$$\text{where:} \quad (133)$$

$$E = \frac{1}{2}kA^2 \quad (134)$$

$$K = \frac{1}{2}mv_x^2 \quad (135)$$

$$U = \frac{1}{2}kx^2 \quad (136)$$

$$\text{thus:} \quad (137)$$

$$K = \frac{1}{2}kA^2 + \frac{1}{2}kx^2 \quad (138)$$

$$= \frac{1}{2}k(A^2 - x^2) \quad (139)$$

$$= \frac{1}{2} \times 200 \frac{\text{N}}{\text{m}} \times [(0.050 \text{ m})^2 - (0.030 \text{ m})^2] \quad (140)$$

$$= 100 \frac{\text{N}}{\text{m}} \times [0.0025 \text{ m}^2 - 0.0009 \text{ m}^2] \quad (141)$$

$$= 100 \frac{\text{kg} \cdot \text{m} \cdot \text{s}^{-2}}{\text{m}} \times (0.0016 \text{ m}^2) \quad (142)$$

$$= 100 \text{ kg} \cdot \text{s}^{-2} \times 0.0016 \text{ m}^2 \quad (143)$$

$$= 0.16 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2} \quad (144)$$

$$= 0.16 \text{ kg} \cdot \frac{\text{m}^2}{\text{s}^2} = 0.16 \text{ J} \quad (145)$$

Thus, at displacement $x = 0.030 \text{ m}$, the kinetic energy is $K = 0.16 \text{ J}$.

```
[12]: k = 200.0 ## N / m
      ## (kg * m * s^-2) / m
      ## kg * s^-2
A = 0.050 ## m
x = 0.030 ## m
tmp = A**2 - x**2
print(f" A          = {A:.14g} m\n"
      f"      x          = {x:.14g} m\n"
      f"    A^2          = {A**2:.14g} m^2\n"
      f"      x^2        = {x**2:.14g} m^2\n"
      f"    (A^2 - x^2) = {tmp:.14g} m^2\n")

E = 1/2 * k * A**2
U = 1/2 * k * x**2
print(f"E = {E:.14g} J\n"
      f"U = {U:.14g} J\n"
      f"K = {E - U:.14g} J\n")
```

$$\begin{aligned}
A &= 0.05 \text{ m} \\
x &= 0.03 \text{ m} \\
A^2 &= 0.0025 \text{ m}^2 \\
x^2 &= 0.0009 \text{ m}^2 \\
(A^2 - x^2) &= 0.0016 \text{ m}^2
\end{aligned}$$

$$\begin{aligned}
E &= 0.25 \text{ J} \\
U &= 0.09 \text{ J} \\
K &= 0.16 \text{ J}
\end{aligned}$$

14. A mass $m = 0.40 \text{ kg}$ hung on a vertical spring is stretched $L = 0.098 \text{ m}$ to reach equilibrium. If the mass is pulled down slightly and released, what is the period of oscillation?

This is a *Vertical SHM* problem. To predict the period of oscillation t , we first need the spring force constant k .

$$k = \frac{mg}{L} \quad (146)$$

$$= \frac{0.40 \text{ kg} \times 9.81 \frac{\text{m}}{\text{s}^2}}{0.098 \text{ m}} \quad (147)$$

$$= 4.0 \frac{\text{kg}}{\text{s}^2} \quad (148)$$

```
[13]: m = 0.40 ## kg
      g = 9.81 ## m / s^2
      L = 0.098 ## m
      k = (m * g) / L

      print(f"m = {m:.14g} kg\n"
            f"g = {g:.14g} m / s^2\n"
            f"L = {L:.14g} m\n"
            f"k = {k} kg / s^2")
```

$$\begin{aligned}
m &= 0.4 \text{ kg} \\
g &= 9.81 \text{ m} / \text{s}^2 \\
L &= 0.098 \text{ m} \\
k &= 40.04081632653062 \text{ kg} / \text{s}^2
\end{aligned}$$

With the spring force constant k , we can predict the period of oscillation t .

$$\omega = \sqrt{\frac{k}{m}} \quad (149)$$

$$= \sqrt{\frac{0.098 \frac{\text{kg}}{\text{s}^2}}{0.40 \text{ kg}}} \quad (150)$$

$$= \sqrt{\frac{0.098 \frac{1}{\text{s}^2}}{0.40}} \quad (151)$$

$$= \sqrt{100.102 \frac{1}{\text{s}^2}} \quad (152)$$

$$= 10.01 \frac{1}{\text{s}} = 10.01 \frac{\text{rad}}{\text{s}} \quad (153)$$

$$\text{thus:} \quad (154)$$

$$\omega = 2\pi f \quad (155)$$

$$f = \frac{\omega}{2\pi} \quad (156)$$

$$= \frac{10.01 \frac{\text{rad}}{\text{s}}}{2\pi} \quad (157)$$

$$= 1.59 \text{ Hz} \quad (158)$$

$$\text{thus:} \quad (159)$$

$$T = \frac{1}{f} \quad (160)$$

$$= \frac{1}{1.59 \text{ Hz}} \quad (161)$$

$$= 0.628 \text{ s} \quad (162)$$

```
[14]: tmp = k / m
print(f"k                = {k:.14g} kg / s^2\n"
      f"      m          = {m:.14g} kg\n"
      f"k / m          = {tmp:.14g} s^-2")

omega = math.sqrt(tmp)
print(f"sqrt(k / m)      = {omega:.14g} rad / s")

f = omega / math.tau
print(f"omega / (2 * math.pi) = {f:.14g} Hz")

T = 1 / f
print(f"T = {T} s")
```

```
k                = 40.040816326531 kg / s^2
      m          = 0.4 kg
k / m            = 100.10204081633 s^-2
```

```

sqrt(k / m)          = 10.005100739939 rad / s
omega / (2 * math.pi) = 1.5923612388937 Hz
T = 0.6279982051652924 s

```

15. A watch's balance wheel has moment of inertia $I = 2.0 \times 10^{-6} \text{ kg} \cdot \text{m}^2$ and its spring has torsion constant $\kappa = 8.0 \times 10^{-4} \text{ N} \cdot \frac{\text{m}}{\text{rad}}$. What is the period of oscillation?

This is an *Angular SHM* problem, specifically dealing with *torsional oscillators*. Recall that this is the rotational analog of Hooke's law, i.e. $F_x = -kx$, where it instead produces *angular* simple harmonic motion. Every SHM formula from *Key Equations of SHM* has a rotational counterpart, so you just swap out the following:

$$\text{Displacement } x \rightarrow \text{Angle } \theta \quad (163)$$

$$\text{Mass } m \rightarrow \text{Moment of Inertia } I \quad (164)$$

$$\text{Spring force constant } k \rightarrow \text{Torsion constant } \kappa \quad (165)$$

In this question, since we have $k = \kappa$ and $m = I$, we can calculate the period of oscillation T in terms of the angular frequency ω .

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{\kappa}{I}} \quad (166)$$

$$\omega = 2\pi f \quad (167)$$

$$f = \frac{\omega}{2\pi} \quad (168)$$

$$\text{since:} \quad (169)$$

$$T = \frac{1}{f} \quad (170)$$

$$\text{thus:} \quad (171)$$

$$T = \frac{1}{\omega} \quad (172)$$

$$= \frac{2\pi}{\omega} \quad (173)$$

$$= \frac{2\pi}{\sqrt{\kappa}} \quad (174)$$

$$= \frac{2\pi}{\sqrt{\frac{\kappa}{I}}} \quad (175)$$

$$= \frac{2\pi\sqrt{I}}{\sqrt{\kappa}} \quad (176)$$

$$= \frac{2\pi\sqrt{2.0 \times 10^{-6} \text{ kg} \cdot \text{m}^2}}{\sqrt{8.0 \times 10^{-4} \text{ kg} \cdot \text{m} \cdot \text{s}^{-2} \cdot \frac{\text{m}}{\text{rad}}}} \quad (177)$$

$$= 0.314 \text{ s} \quad (178)$$

```
[15]: I      = 2.0e-6 ## kg / m^2
kappa = 8.0e-4 ## N * (m / rad)
T      = math.tau * math.sqrt(I) / math.sqrt(kappa)

print(f"I      = {I:.14g} kg / m^2\n"
      f"kappa = {kappa:.14g} M * (m / rad)\n"
      f"T      = {T:.14g} s")
```

```
I      = 2e-06 kg / m^2
kappa = 0.0008 M * (m / rad)
T      = 0.31415926535898 s
```

16. Why can small vibrations of two bonded atoms about their equilibrium separation R_0 be treated as simple harmonic motion (SHM)?

The potential energy $U(r)$ near the minimum equilibrium point at distance $r \approx R_0$ is approximately

17. The bob of a simple pendulum is replaced with one twice as heavy (i.e. $m' = 2m$), and the swing angle is increased from 5 degrees to 10 degrees.

Recall the formula for the period T of a *simple pendulum*:

$$T = 2\pi\sqrt{\frac{L}{g}} \quad (179)$$

Since neither mass m nor swing angle θ are involved in the formula, the period remains unchanged.

18. What length must a simple pendulum have to give a period of $t = 3.00$ s?

We can use the formula for the period T of a simple pendulum to isolate the length L .

$$T = 2\pi\sqrt{\frac{L}{g}} \quad (180)$$

$$T = 2\pi\frac{\sqrt{L}}{\sqrt{g}} \quad (181)$$

$$\frac{\sqrt{g}}{2\pi} \times T = \sqrt{L} \quad (182)$$

$$L = \left(T\frac{\sqrt{g}}{2\pi}\right)^2 \quad (183)$$

$$= T^2\frac{g}{4\pi^2} \quad (184)$$

$$= (3.00 \text{ s})^2 \times \frac{9.81 \frac{\text{m}}{\text{s}^2}}{4\pi^2} \quad (185)$$

$$= 9.00 \text{ s}^2 \times 0.2485 \frac{\text{m}}{\text{s}^2} \quad (186)$$

$$= 2.23 \text{ m} \quad (187)$$

```
[16]: T = 3.00 ## s
      g = 9.81

      L = T**2 * (g / (4 * math.pi**2))
      print(f"T = {T} s\n"
            f"g = {g} m / s^2\n"
            f"L = {L} m")
```

```
T = 3.0 s
g = 9.81 m / s^2
L = 2.236411825945501 m
```

19. A simple pendulum-clock is moved to the Moon, where the acceleration constant of gravity g is one-sixth of its value on Earth. Its period....

Recall the formula for the period T of a simple pendulum:

$$T = 2\pi\sqrt{\frac{L}{g}} \quad (188)$$

$$T = 2\pi\sqrt{\frac{L}{6g}} \quad (189)$$

$$(190)$$

Since g is inversely proportional to the period T , the period is decreased by a factor of the square root of the multiplier, i.e. $\sqrt{6}$.

20. A uniform meter stick with length ($L = 1.00$ m) is pivoted at one (1) end and swings as a physical pendulum. For a rod about its end, the moment of inertia is $I = \frac{1}{3}mL^2$. What is its period for small swings?

Recall that the moment of inertia I is an analog to mass m in simple harmonic motion. Since we have that, we can use it to calculate

$$T = 2\pi\sqrt{\frac{L}{g}} \quad (191)$$

```
[17]: math.tau * math.sqrt(1/3 / 9.81)
```

```
[17]: 1.1582031381206312
```

21. As the damping constant b of an oscillator is increased (still underdamped), what happens?

The amplitude decays faster and the period increases slightly.

22. A damped oscillator has a mass $m = 0.50$ kg and damping constant $b = 0.10 \frac{\text{kg}}{\text{s}}$. Its amplitude envelope $A_e = Ae^{-bt/2m}$. How long does it take for the amplitude to fall to $1/e$ of its initial value?

$$A_e = Ae^{-(b/2m)t} \quad (192)$$

$$\frac{1}{\frac{e}{A}} = \frac{Ae^{-bt/2m}}{A} \quad (193)$$

$$\frac{A}{e} = e^{-bt/2m} \quad (194)$$

$$\text{where:} \quad (195)$$

$$b^y = x \rightarrow \log_b(x) = y \quad (196)$$

$$\text{thus:} \quad (197)$$

$$e^{-bt/2m} = \frac{A}{e} \quad (198)$$

$$\log_e\left(\frac{A}{e}\right) = -\frac{bt}{2m} \quad (199)$$

$$\log_e(A) - \log_e(e) = -\frac{bt}{2m} \quad (200)$$

$$\log_e(A) - 1 = -\frac{bt}{2m} \quad (201)$$

$$\log_e \frac{1}{e} - 1 = -\frac{bt}{2m} \quad (202)$$

$$\log_e(1) - \log_e e - 1 = -\frac{bt}{2m} \quad (203)$$

$$0 - 1 - 1 = -\frac{bt}{2m} \quad (204)$$

$$-2 = -\frac{bt}{2m} \quad (205)$$

$$-\frac{2m}{b} \times -2 = -\frac{bt}{2m} \times -\frac{2m}{b} \quad (206)$$

$$t = \frac{4m}{b} \quad (207)$$

```
[1]: print(4*.50/.10)
```

20.0

24. A driven oscillator has mass $m = 1.0$ kg, restoring force constant $k = 100 \frac{\text{N}}{\text{m}}$, damping constant $b = 2.0 \frac{\text{kg}}{\text{s}}$ and maximum force $F_{max} = 5.0$ N. It is driven at $\omega_d = 10 \frac{\text{rad}}{\text{s}}$. What is the steady-state amplitude?